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Introduction to Statistics

By Simanga Mchunu | 2026

https://introductiontostats.vercel.app/

Agenda

- What is Statistics?
- What is Data?
- Types of Statistics
- Descriptive Statistics Essentials
- Inferential Statistics
- Data Distributions
- Probability
- Case Study: Predicting Students Success

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01 - Introduction to Statistics

Foundations for Data Science · 2026

What is Statistics?

- Science of collecting, analyzing, interpreting, presenting, and organizing **data**.
- Used in business, healthcare, sports, and machine learning.
- It is **making sense** of numbers.
- Making **informed decisions** in the presence of uncertainty and variation.
- Organising and summarizing **data** to **draw conclusions** based on the information contained in the data.

What is Data?

Data is **information collected through observation, measurement, or facts**, used for **reference, analysis, or research**.

It represents raw inputs that, when structured and interpreted, form the basis of knowledge and decision-making

Types of Statistics

- *“The numbers have no way of speaking for themselves. We speak for them. We imbue them with meaning". - Nate Silver, The Signal and the Noise*

Descriptive Statistics:

Summarize data (mean, median, variance).

Inferential Statistics:

Draw conclusions (hypothesis testing, confidence intervals).

Descriptive Statistics Essentials

Mean: Average = $\mu = (1/n) \sum x_i$)

Median: Middle value

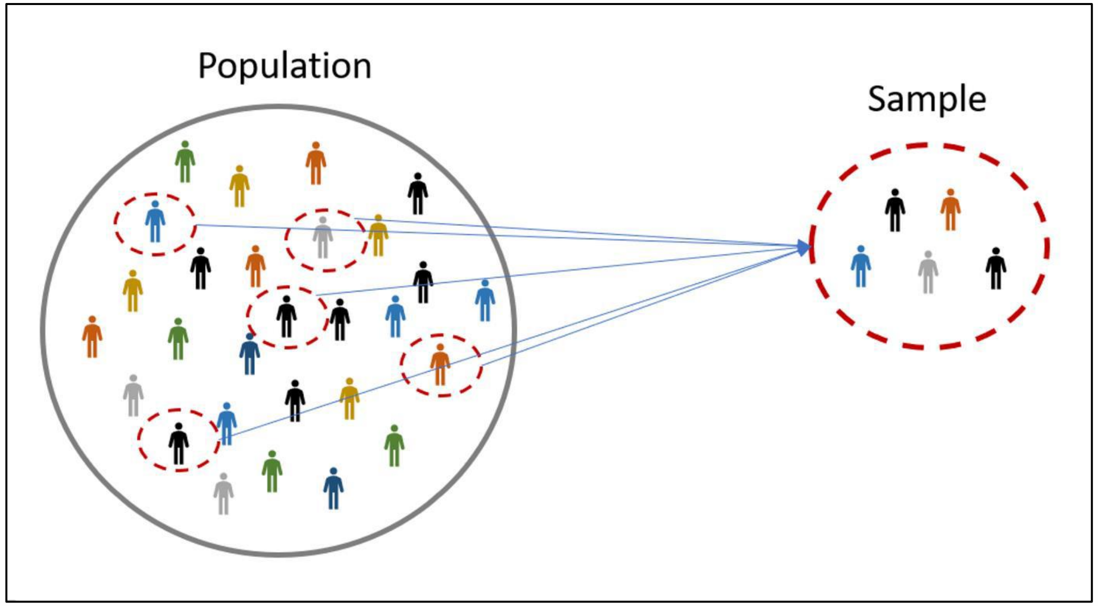
Mode: Most frequent value

Variance: Spread = $\sigma^2 = (1/n) \sum(x_i - \mu)^2$

Standard Deviation = $\sqrt{\text{Variance}}$

Inferential Statistics Concepts

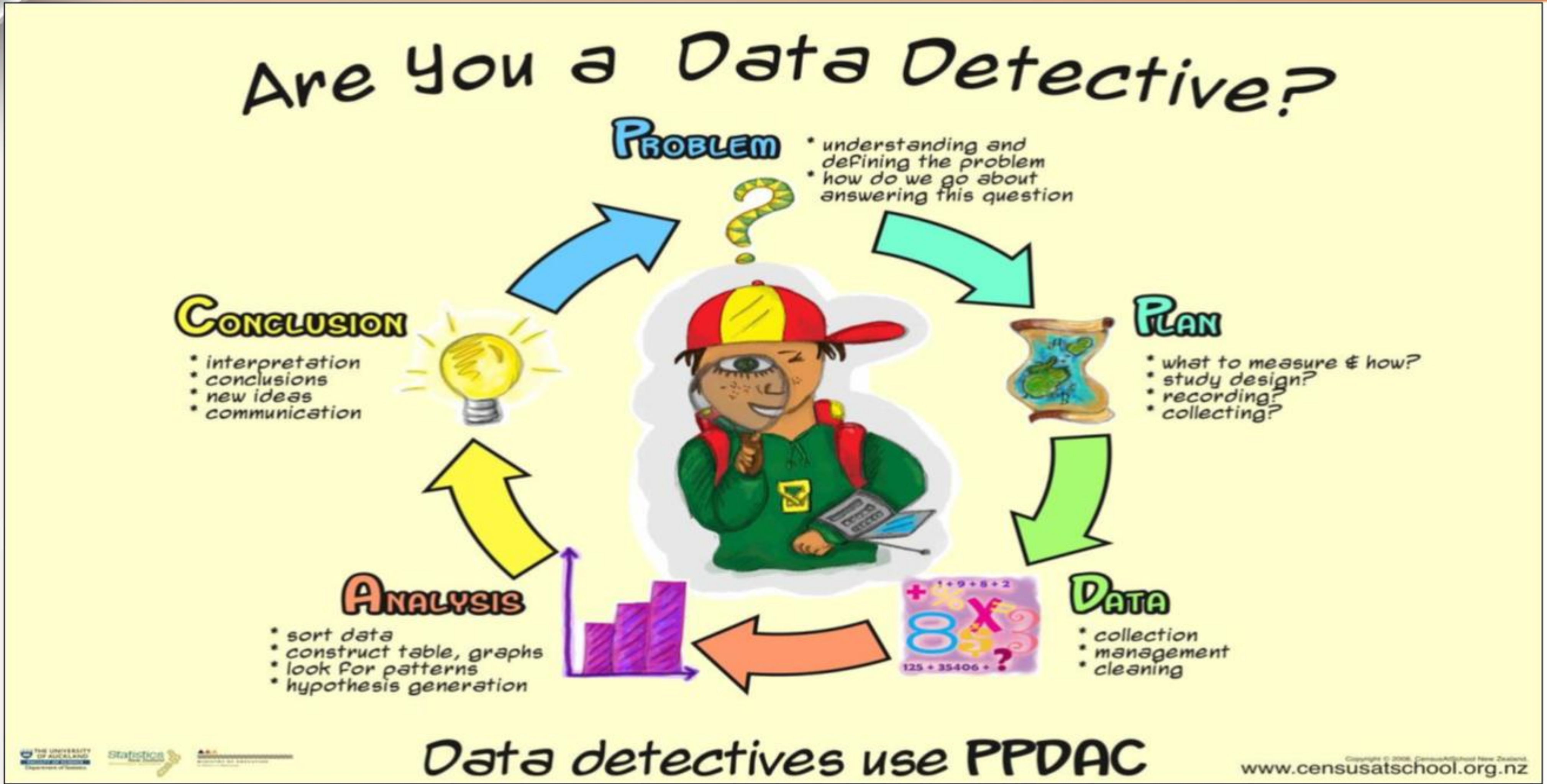
- **Sample** vs. **Population**
- Confidence Intervals
- Hypothesis Testing: Null & Alternative Hypotheses, p-values
- **Population:** Collection of objects about which information is sought.
- **Sample:** Part of the population that is observed.
- **Spiegelhalter's key idea:** Even when you seem to have all the data, treat it as a sample from a metaphorical population , the imaginary space of all possible observations that could have arisen. This means every finding is an estimate, not a fact.
- **p-values warning:** Spiegelhalter is emphatic that many studies claiming statistical significance are misleading. A p-value below 0.05 does not mean a result is true , it means the data would be unlikely if the null hypothesis were true. Small samples amplify this problem.



Source [1]: https://medium.com/analytics vidhya/population sample parameter statistic biased unbiased ead2021d93d7

Source [2]: https://www.lse.ac.uk/Events/2019/03/20190327t1830vHKT/Learning-from-Data

Data Distributions



Shipman Murdered More Than 200 Patients, Inquiry Finds

The British former GP Harold Shipman murdered at least 215 of his patients, the first phase of the public inquiry into the serial killings concluded last week. There is a "real suspicion" that he claimed the lives of another 45 victims, according to the judge leading the inquiry.

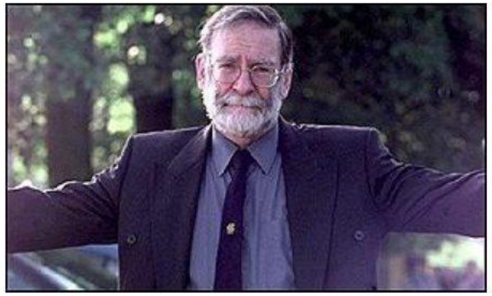
High court judge Dame Janet Smith said Shipman may have been hoping to get caught when he altered the will of his last victim, Kathleen Grundy, 81. The "crude forgery" of the will "made detection inevitable," she said, concluding: "It is hard to resist the inference that Shipman was driven by a need to draw attention to himself and his crimes."

The inquiry examined a total of 888 cases in its 2000 page report, *Death Disguised*. Shipman was found not responsible for 604 deaths, and no conclusion was reached in a further 38 cases. The 215 people that the inquiry determined were killed by Shipman comprised 171 women and 44 men.

Source [3]: https://pmc.ncbi.nlm.nih.gov/articles/PMC1123718/

Looking at Data: The Shipman Case

- What was the pattern of Harold Shipman's murders?
- **Problem**: can more detail tell us more about what Shipman did?
- **Plan**: compare actual times at which his patients died with the times of deaths recorded by other local GPs
- **Data**: a huge exercise requiring examination of death certificates
- **Analysis**: simple plotting of death times revealed a stark afternoon cluster
- **Conclusion**: the pattern was statistically impossible by chance — but required domain context to interpret correctly



'I have nothing to hide'

Dr Harold Shipman, general practitioner, on his arrest in September 1998

Source [4]: The art of Statistics by David Spiegelhalter's book

- **Critical lesson:** The same test applied to all GPs surfaced a second outlier with an entirely innocent explanation (elderly rural patients). Without domain context, statistics produces dangerous conclusions. This is PPDAC — see next slide.

The PPDAC Cycle A Framework for Every Statistical Investigation

- **PPDAC** is the named framework Spiegelhalter uses throughout The Art of Statistics. Every investigation follows fivestages in order.

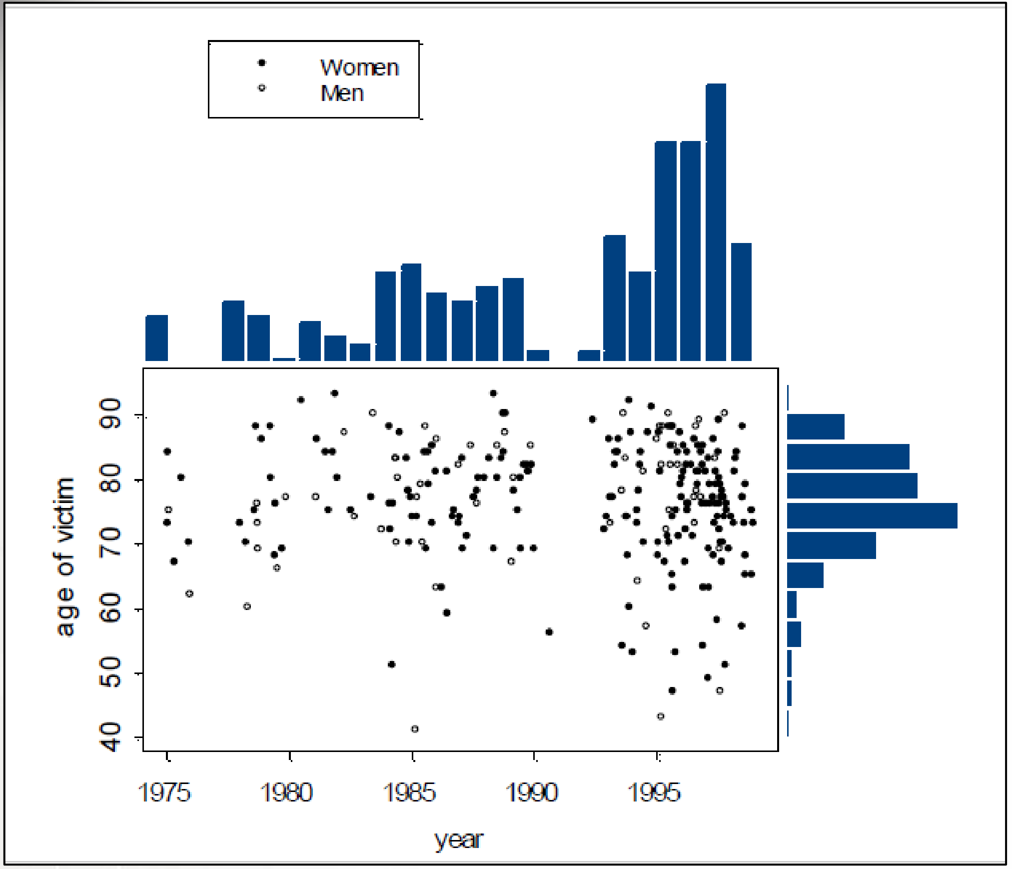
- **P — Problem:** State the question precisely before touching data. A vague question produces a useless answer.
- **P — Plan:** Decide what to measure, from whom, and how. Sampling design matters enormously.
- **D — Data:** Collect, inspect, and understand limitations. What is missing? How was it recorded?
- **A — Analysis:** Apply statistical methods. Summaries, visualisations, models — always with uncertainty.
- **C — Conclusion:** Return to the Problem. Translate findings into an honest answer. State what the data cannot tell you.

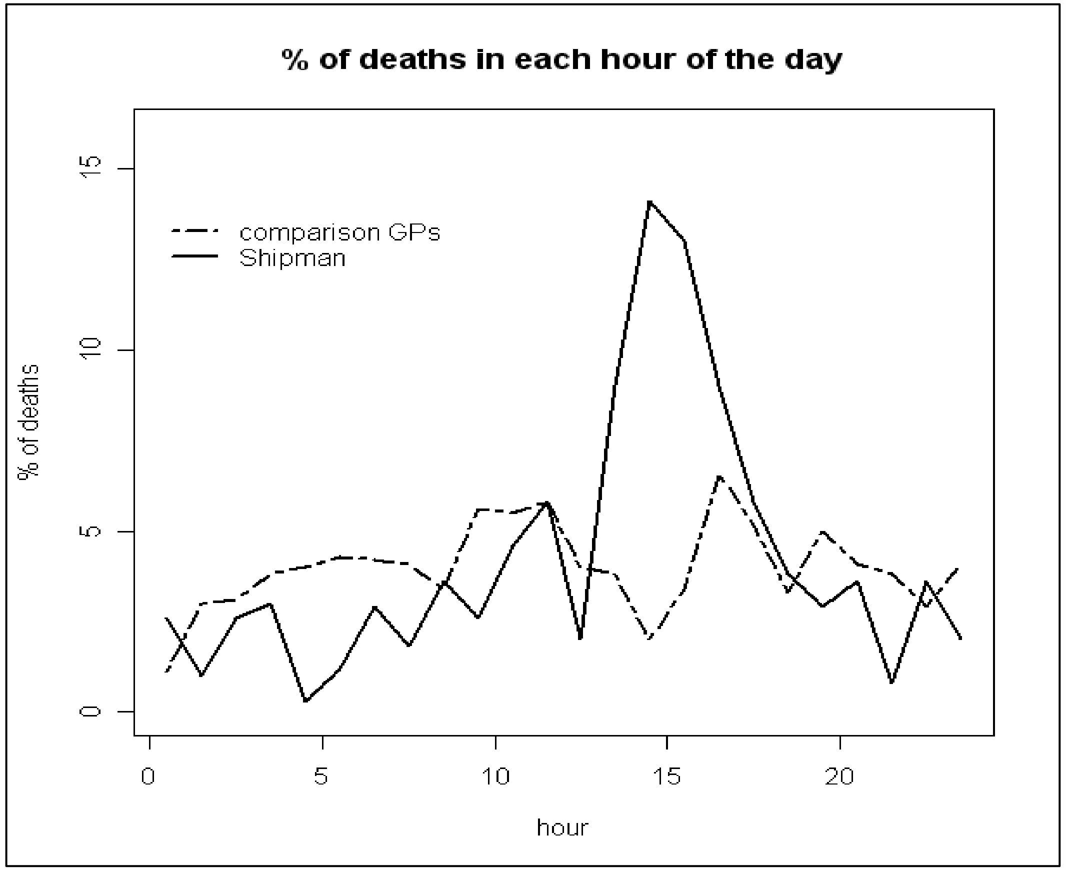
- The Shipman case (previous slide) followed PPDAC exactly: the Problem drove every decision about which data to collect and how toanalyse it. Without that structure, the pattern would never have been found.

- **Warning: most students jump directly to Analysis. This is how wrong conclusions are reached.**

Source [4]: The Art of Statistics by David Spiegelhalter

Looking at Data

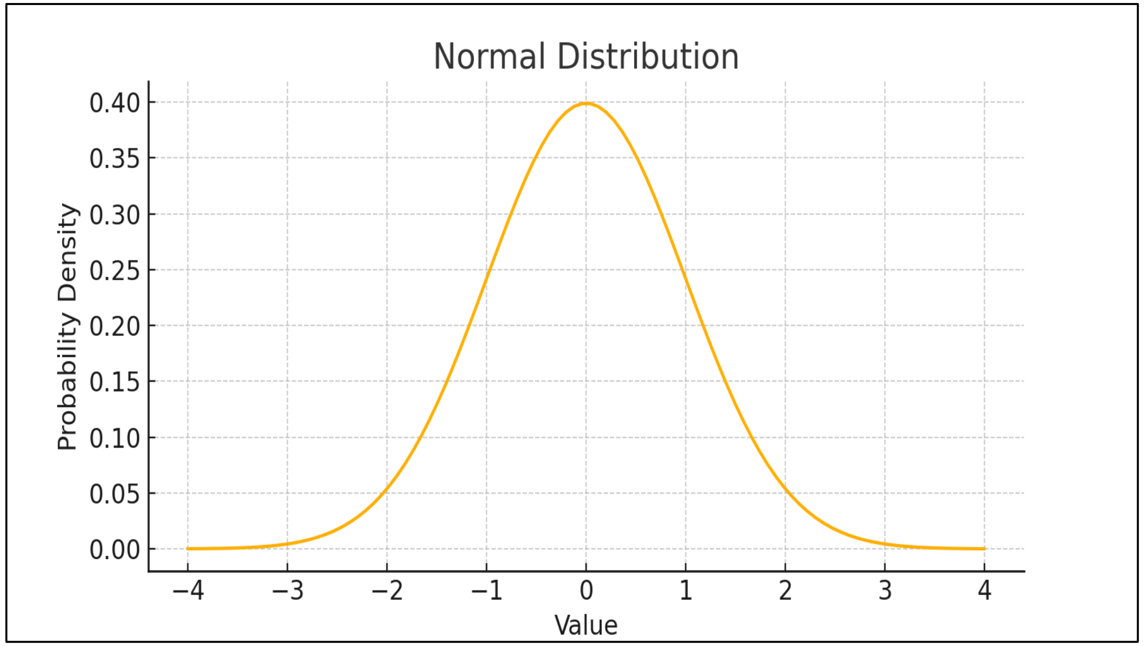




Source: [4] https://www.lse.ac.uk/Events/2019/03/20190327t1830vHKT/Learning-from-Data

What is a Data Distribution?

- Normal Distribution (Known as the **Gaussian Distribution**) is a probability distribution that appears as a "bell curve" when graphed. Also called the **Bell Curve** because of its shape.
- **Characteristics**:
  - **Symmetrical** around the mean
  - Mean = Median = Mode
  - Most data falls near the center
  - Tails taper off equally on both sides
  - Defined mathematically by the **Gaussian function**



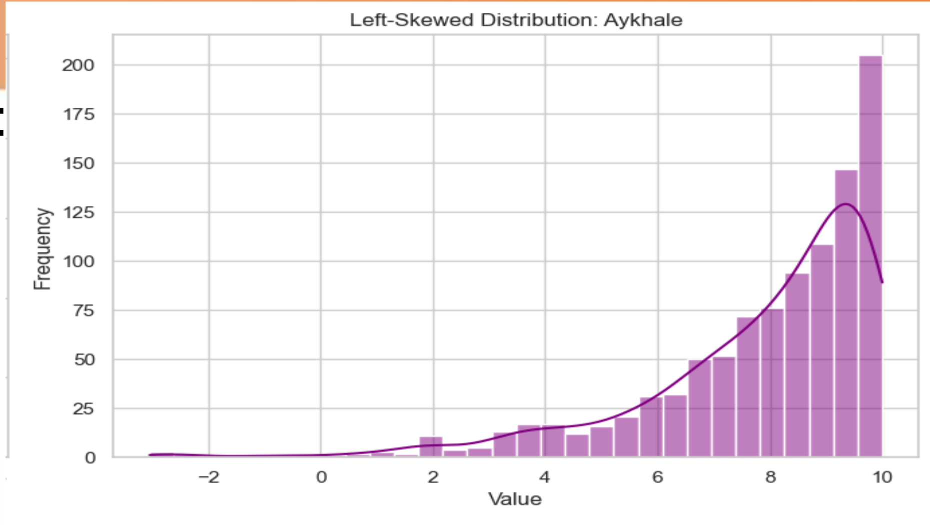
What is Skewed Distribution?

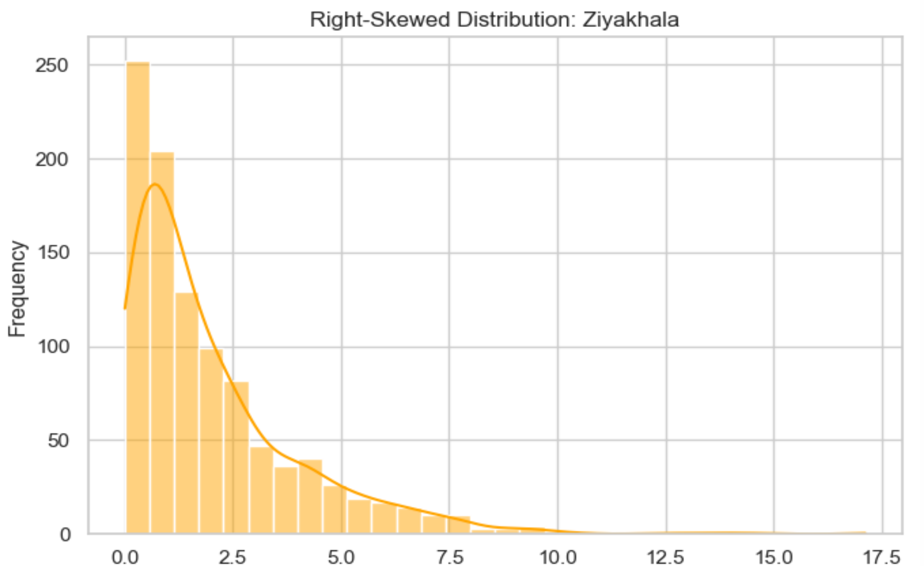
A **skewed distribution** occurs when the **data is not evenly distributed** around the mean. The values tend to **lean more to one side**, forming a **tail** on either the right or left.

Characteristics:

- **Asymmetrical** shape
- The **mean, median, and mode are not equal**
- **Two types** of skewness:
  - **Right-Skewed (Positive Skew)**: Tail on the **right**, Mean > Median > Mode
  - **Left-Skewed (Negative Skew)**:
  - Tail on the **left**, Mean < Median < Mode

Affects **statistical analysis and interpretation**



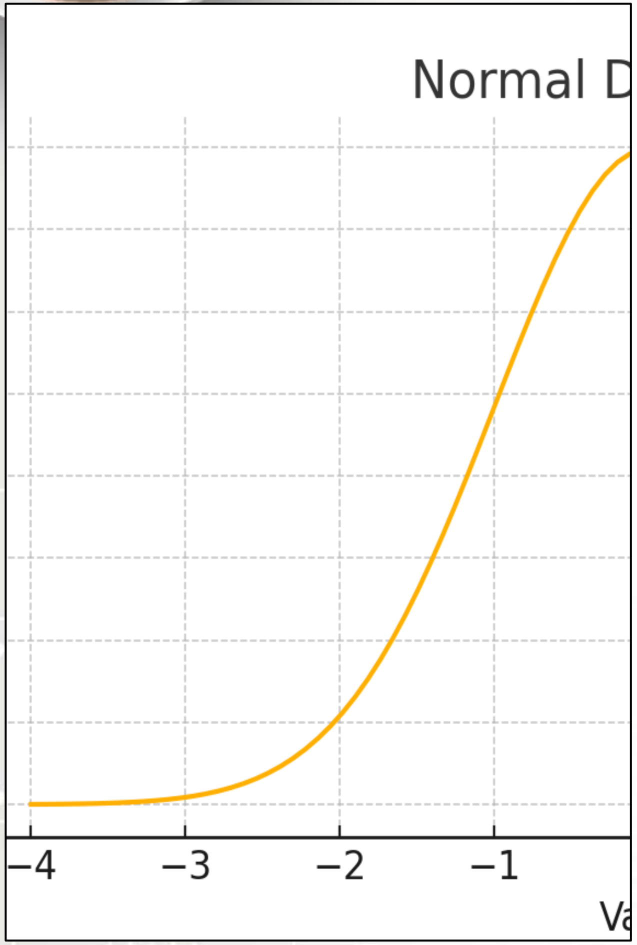


Which Central Tendency Measure to Choose?

Measure	Best When	Avoid When
Mean	Data is evenly distributed (normal distribution). No outliers	Sensitive to outliers
Median	Data has outliers or skewed distribution	Less informative for symmetric, well-behaved data
Mode	Data is categorical or you need the most frequent value	Not very useful for continuous data or when no clear mode exists

In probability theory, the **central limit theorem** (**CLT**) states that, under appropriate conditions, the distribution of a normalized version of the sample mean converges to a standard normal distribution.

What Distribution Means in Data Science



Understanding if your data is **normally distributed** matters because:

Many algorithms assume it:

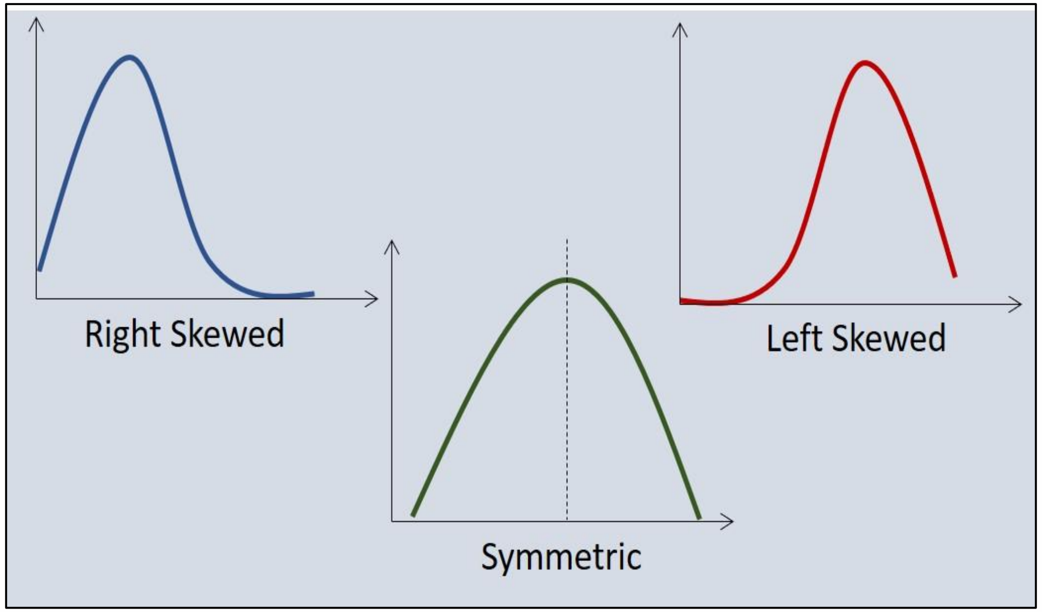
- Linear regression
- Logistic regression
- Naive Bayes
- T-tests and ANOVA

Standardized models:

- Let you apply **z-scores** (how many std deviations from mean)
- Make it easier to detect **outliers**

What Distribution Means in Data Science

- **Helps with:**
  - Feature scaling
  - Probability estimates
  - Statistical inference



- **Real-World Example:**
  - If you're analyzing **student test scores**, a normal distribution means:
    - Most students score near the average
    - Fewer students score very high or very low
    - The grading curve might be centered on the class average

Probability: The Language of Uncertainty

- **Formula for an event A:**

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

- **Why it matters in Data Science:**
  - Handles **uncertainty** in predictions (e.g., spam detection, fraud detection)
  - Used in probabilistic models like **Naive Bayes**, **Bayesian networks**, **Hidden Markov Models**
  - Foundation of **sampling**, **A/B testing**, and **predictive modeling**

Probability Example

- If 3 out of 5 emails are spam:

$$P(\text{Spam}) = \frac{3}{5} = 0.6$$

Conditional Probability: Knowing Changes the Odds

- **Formula:**

$$P(A|B) = P(A \cap B) / P(B)$$

- **Data Science Example:**
  - What's the chance a user **buys** given they **clicked**?
  - Let A = Buy, B = Click
  - Crucial for **recommendation systems** and **conversion analysis**

Bayes' Theorem: Update Your Beliefs

Formula:
P(A|B) = P(B|A) * P(A) / P(B)

Logic:
Combine prior knowledge P(A) and observed data P(B|A) to **update belief**

Core to **Bayesian reasoning**, **probabilistic classifiers**, and **medical diagnostics**

Bayesian Thinking: Updating Beliefs With Evidence

- Bayes' theorem is not just a formula. It encodes a way of thinking that Spiegelhalter considers foundational to all statistics.

- **The core idea : three steps:**
- **1. Prior:** Start with a belief. What do we think before seeing any data?
- **2. Likelihood:** Collect evidence. How probable is this evidence if our belief is true?
- **3. Posterior:** Update the belief. P(Hypothesis | Evidence) using Bayes' rule.

- **Worked Example Shipman revisited:**
- Prior: a GP has a normal rate of patient deaths. Likelihood: Shipman's death rate among afternoon patients is vastly higher than expected. Posterior: after updating on this evidence, the probability that his pattern is due to chance becomes vanishingly small.

- This is why Bayesian thinking matters: we do not need to wait for a definitive test. We update our confidence continuously as evidence accumulates.

- In data science: spam filters, medical diagnosis, fraud detection and recommendation engines all use Bayesian updating at their core.

Source [4]: The Art of Statistics : Chapter 8: Probability, the Language of Uncertainty

Data Science Use Case: Naive Bayes Classifier

- Data Science Use Case – Naive Bayes Classifier:
  - Let's classify if a message is **spam** or **not spam**:
    - A: Message is spam
    - B: Message contains the word "free“

$$P(Spam|\text{"}free\text{"}) = \frac{P(\text{"}free\text{"}|Spam) \cdot P(Spam)}{P(\text{"}free\text{"})}$$

- Trained on labelled data to estimate all components.

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Case Study: Predicting Student Success

Applying the PPDAC workflow end-to-end

Case Study:Predicting Student Success

Practical Case Study

Scenario: A university wants to identify students at risk of failing beforethe final exam.

Your dataset:

Student 1 — 5 hrs study, 60% attendance, 45 assignment, 48 exam

Student 2 — 8 hrs study, 75% attendance, 58 assignment, 60 exam

Student 3 — 10 hrs study, 80% attendance, 62 assignment, 67 exam

Student 4 — 12 hrs study, 85% attendance, 70 assignment, 74 exam

Student 5 — 15 hrs study, 90% attendance, 82 assignment, 85 exam

Student 6 — 18 hrs study, 95% attendance, 90 assignment, 92 exam

Variables: Study Hours | Attendance | Assignment Score | Exam Score

Steps 1 & 2: Understand the Problem and Explore the Data

Step 1: Understand the Problem

Before calculating anything - ask questions:

- What does success mean in this context?
- Which variables might influence performance?
- What is missing from the data?

Statistical Thinking: The question comes before the data.

Step 2: Explore the Data

Calculate summary statistics:

- Mean study hours — what is typical?
- Mean attendance percentage
- Mean assignment and exam scores

Discussion: Are there any unusual observations?

Step 3: Measure Variability (The Spread of the Data)

Why does variability matter?

Two datasets can have the same mean but be completely different.

For Study Hours and Final Exam Scores, calculate:

- Range = Maximum value − Minimum value
- Variance = Average squared distance from the mean
- Standard Deviation = Square root of variance (typical distance from the mean)

Example — Study Hours:

- Values: 5, 8, 10, 12, 15, 18
- Mean = 11.33 | Range = 13
- Standard Deviation ≈ 4.5 → Data is moderately spread

Key Insight: High variability = students are very different from each other.

Step 4: Visualise the Data (See Before You Model)

Histogram of Final Exam Scores:

- Is the distribution symmetric or skewed?
- Are scores clustered together or spread out?
- Are any scores unusually low (potential at-risk students)?

Scatter Plot — Study Hours vs Final Exam Score:

- Plot each student as a point: x = study hours, y = exam score
- Look for a trend: do more study hours lead to higher scores?
- Identify outliers: students who studied a lot but still scored low

Golden Rule of Data Science:

Always visualise before you model. A chart reveals what a formula cannot.

Visualisation reveals patterns, clusters, outliers, and errors.

Step 5: Investigate Relationships (Correlation)

Correlation tells us: do two variables move together?

Calculate correlation between:

- Study Hours vs Exam Score
- Attendance vs Exam Score
- Assignment Score vs Exam Score

Interpreting the correlation value (r):

- r close to +1 → Strong positive relationship
- r close to 0 → Weak or no relationship
- r close to −1 → Strong negative relationship

Warning - Correlation is NOT Causation:

Two variables can move together without one causing the other.

Always look for a logical reason behind the relationship.

Step 6: Build a Simple Statistical Model (No ML Needed)

Using the scatter plot, draw an estimated line of best fit.

Linear Relationship Formula:

Exam Score ≈ a + b × Study Hours

From our data — estimated values:

- Intercept (a) ≈ 22 | Slope (b) ≈ 4
- So: Exam Score ≈ 22 + 4 × Study Hours

What the slope means:

For every additional hour of study per week,

exam performance increases by approximately 4 marks.

This is regression — the foundation of predictive modelling.

We reached prediction using only statistics.

Steps 7 & 8: Make a Prediction and Quantify Uncertainty

Step 7: Use your model to make a prediction.

Scenario: A new student studies 13 hours per week.

Using our formula:

Predicted Exam Score = 22 + 4 × 13 = 74 marks

Step 8: Can we be certain about this prediction?

No. Statistics gives us estimates, not guarantees.

Reasons predictions can fail:

- Motivation and stress levels are not in the data
- Health, family circumstances, or external disruptions
- Sample is small (only 6 students)

Key Principle: Always report your uncertainty alongside your prediction.

Step 9: Statistical Decision Making

Scenario: The university can only support 20 at-risk students.

How do you decide who needs help most?

Use evidence from multiple variables:

- Low study hours (below 8 hrs per week)
- Low attendance (below 70%)
- Low assignment score (below 55)

Statistical Approach:

- Rank students by predicted exam score
- Identify those predicted to score below 50%
- Investigate further: is the risk consistent across variables?

Important: Every recommendation must be backed by evidence.

Opinion without data is just guessing.

Step 10: Final Reflection — The Full Statistical Workflow

You have just completed the process used by:

- Data Scientists
- Public Health Researchers
- Business Analysts
- Economists
- AI Engineers

The complete workflow:

Question → Data → Exploration → Visualisation

→ Correlation → Model → Prediction → Decision

What is remarkable:

We reached prediction and decision-making without using machine learning.

We used only statistical thinking.

The most important skill in data science is not knowing algorithms —

it is knowing how to ask the right question and reason from evidence.

Can 6 Students Represent a Population? The Critical Question

- **Spiegelhalter insists: before generalising any finding, ask; what population does this sample represent, and how well?**
- Our case study dataset: 6 students. University population: potentially thousands.
- **Five questions every analyst must answer:**
- 1. Were these 6 students randomly selected? If not, the sample may be biased ; volunteers tend to be more motivated than average.
- 2. What is the target population? This university? All South African students? The answer changes whether findings generalise.
- 3. What cases did we not observe? Students who study 0 hours, students who fail despite high attendance ; our model has never seen these.
- 4. What variables are entirely absent? Socio economic background, part-time employment, access to resources, language of instruction.
- 5. Is our correlation real or a small-sample artefact? With n=6, extreme r values can appear by chance even when no population-level relationship exists.
- Spiegelhalter's metaphorical population: treat results as a sample from the imaginary space of all students who could have been in this situation. Our findings are always estimates, never certainties.

Source [4]: The Art of Statistics — Chapter 3: Why Are We Looking at Data Anyway? Populations and Measurement

Ethical Responsibility: What the Analyst Owes the Student

- **We labelled Student 1 “high risk.” This has real consequences for a real person. Statistics without ethics is dangerous.**
- **Four obligations Spiegelhalter places on every analyst:**
- **1. Report uncertainty alongside every estimate.** “Student 1 may fail” is irresponsible. “Predicted score: 30.4, typical error ±3.2 marks, based on 6 students” is honest.
- **2. State what the data cannot support.** Attendance and study hours are associated with performance in this sample ; we cannot claim causation, and we cannot claim this holds for the full student body.
- **3. Consider who is harmed by errors.** A false positive (wrongly labelled at-risk) may stigmatise a student. A false negative (missed at-risk) may leave a struggling student without support.
- **4. Ensure decision-makers understand the limits.** If an academic advisor acts on this analysis without understanding its uncertainty and sample size, the analyst shares responsibility for the outcome.

- Spiegelhalter’s closing principle: Statistics produces honest estimates with uncertainty made visible ; so that humans can make better decisions. The moment an analyst forgets this, the numbers become dangerous.

- **Ask yourself: if this analysis was wrong, who would bear the cost? Could that person challenge the conclusion? Do they even know it was made about them?**

Source [4]: The Art of Statistics — Ethical Communication of Statistical Findings

Key Takeaways: What Every Student Must Remember

- **Statistics is the foundation of all data science and AI.**

- Data is NOT reality ; it is only a representation
- Samples represent populations ; never assume they are perfect
- Variability matters as much as averages
- Correlation is NOT causation ; always look for logical reasons
- Every prediction has uncertainty ; always report your confidence
- Evidence must drive every recommendation

- **When you encounter a data challenge:**
- *Do NOT rush to machine learning.*
- First: Understand the problem.
- Then: Explore, visualise, and reason statistically.
- *The best data scientists start with questions, not algorithms.*

Quiz Time!

- **Where the mean, mode, and median are equal? Which graph distribution is:**

A. Negative Skewed
B. Normal Distribution
C. Data Distribution
D. None of the Above

Answer: **B. Normal Distribution**

- **What is Statistics?**

A. Information collected through observation, measurement
B. The state of the country.
C. Science of collecting, analyzing, interpreting, presenting, and organizing data
D. All of the above

Answer: **C. Science of collecting, analyzing, interpreting, presenting, and organizing data**

Quiz Time!

- **What does a skewed distribution indicate?**

  A. Data has multiple peaks

  B. Data tails off more on one side
  C. Data is symmetrical
  D. Data is uniformly distributed
  - **Ans: Correct Answer:** B
- **What does standard deviation measure in a dataset?**

A. The average value of the data

B. The most frequent value

C. The median of the data

D. The spread or variability of the data around the mean

Ans: **Correct Answer:** D

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Any Questions?

Thank you for your attention

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